



Alexis Imbert

PhD Student

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Languages

French: Native
English: B2 - Professional (825/990 TOEIC)
Spanish: B2

Education

Eng	INSA Rouen Normandie , Computer Science and Information Technology	Rouen, France 2021 – Sept 2024
MSc	University of Rouen , Data Science and Engineering	Rouen, France 2023 – June 2024
Exc	Royal Melbourne Institute of Technology (RMIT) , Computer Science • Semester abroad, courses on AI for professionals, ethics in AI	Melbourne, Australia Feb 2023 – July 2023
PEIP	Sorbonne University , General scientific preparatory program	Paris, France 2019 – 2021

Experience

INSA Rouen Normandie , PhD Student, Sulcal Pit Analysis of Cortical Surfaces with GNN Thesis supervised by Paul Honeine (University of Rouen), Guillaume Auzias (Institut de Neurosciences de la Timone, CNRS, Aix-Marseille University), and Benoît Gaüzère (INSA Rouen Normandie). Funded by the French National Research Agency as part of the FAMOUS project. • Hierarchical Information Analysis in Brain Graphs: How can graph machine learning models effectively capture and utilize hierarchical information in brain graphs to enhance the representation and analysis of cortical folding patterns? • Robustness to MRI Acquisition Variations: Can a graph machine learning model trained on brain data from one acquisition center generalize effectively to data from different centers, improving reproducibility in neuroimaging studies? This research explores transfer learning applications in GML. • Local Variation Identification for Cognitive Functions: How can GML methods identify and analyze local variations in brain anatomy, and what insights can these variations provide regarding cognitive functions and neurological conditions? • Publication at GBR 2025.	Rouen, France 2024 – 2027 3 years
Teklia , Research Engineer Internship, Historical Image Description with LLM Automatic description of historical Japanese photography using large multimodal language models (LLM). Project in collaboration with the Guimet Museum of Asian Arts and Teklia. • Benchmarking of prompt engineering software • Image processing for consistency • Large-scale experiment deployment • Fine-tuning of existing models with project images • Poster presentation at SIFED 2024	Paris, France Feb 2024 – Oct 2024 9 months

Skills

Machine Learning: Python, Scikit-learn (contributor), PyTorch, PyTorch-Geometric, Torch-Lightning, LLM, vLLM, YOLO - Ultralytics (contributor), Open-source contributions

Programming: Python, C, Java, Octave & Matlab, HTML, CSS, JavaScript, Node.js, Vue.js, AJAX, Flask, PHP, Django

Sports: Swimming, Climbing, Skiing, Snowboarding

Music: Musical training and classical guitar, solo and in groups for 13 years

Publications

Insights on Using Graph Neural Networks for Sulcal Graphs Predictive Models

June 2025

Alexis Imbert, Benoit Gauzere, Sylvain Takerkart, Guillaume Auzias, and Paul Honeine.

Alexis Imbert, Benoit Gauzere, Sylvain Takerkart, Guillaume Auzias, Paul Honeine

[10.1007/978-3-031-94139-9_8](https://doi.org/10.1007/978-3-031-94139-9_8) (GBR 2025)

Évaluation de Large Language Models pour la description de photographies

June 2024

Alexis Imbert, Mélodie Boillet, and Christopher Kermorvant.

Alexis Imbert, Mélodie Boillet, Christopher Kermorvant

sifed2024.sciencesconf.org/resource/page/id/3 (SIFED 2024)

Projects

INSA Certified Project

2023

Project in collaboration with Actémium Paris Transport. Object detection in images and videos within the compressed domain, applied to road surveillance videos in tunnels.

Web Platform for Academic Mobility Support

2023

Design of a web platform to facilitate administrative procedures for academic mobility from the perspectives of students and faculty.

Fermi and von Neumann Elephant

2022

Project illustrating Von Neumann's elephant concept, with functionalities for generating new designs, computing Fourier series, and reproducing initial designs with adjustable parameters.

"Pixel War" Multiplayer Game

2022

Recreation of the "Pixel War" multiplayer game in a team of three, including backend, frontend, and database development. Players collaboratively place pixels on a canvas to create drawings.

Recognition of Visually Identical Images

2021

Application of mathematical methods to detect pairs of visually identical images, accounting for variations in size, noise, and transformations. Introduction to machine learning concepts.

Spellchecker in C

2021

Team project in C to develop a spellchecker using tree structures.

Volunteer

INSA Rouen Normandie, Representative of the student body at the scientific council of INSA Rouen Normandie.

Rouen, France
Dec 2025 – Nov 2026

Representative of the student body at the scientific council of INSA Rouen Normandie.

Pint of Science, Co-Coordinator of the Rouen 2026 and 2027 edition of the Pint of Science festival.

Rouen, France
Sept 2025 – June 2027

Co-Coordinator of the Rouen 2026 and 2027 edition of the Pint of Science festival.

- Organized and coordinated the Pint of Science festival in Rouen for two consecutive years.
- Managed a team of volunteers and collaborated with local scientists to present engaging talks to the public.

INSA Rouen Normandie, Representative of the student body at the board of directors of INSA Rouen Normandie.

Representative of the student body at the board of directors of INSA Rouen Normandie.

Rouen, France
Dec 2022 – Nov 2024

INSA Rouen Normandie, Representative of the student body at the Department council the ITI (study department) of INSA Rouen Normandie.

Representative of the student body at the Department council the ITI (study department) of INSA Rouen Normandie.

Rouen, France
Oct 2021 – July 2024

INSA Rouen Normandie, President of the Photography Club

President of the Photography Club at INSA Rouen Normandie.

- Coordinated with other student organizations for collaborative events.

Rouen, France
Sept 2023 – Feb 2024

Hélène Boucher High School, President of the Sylver Photography Club

President of the Sylver Photography Club at Hélène Boucher High School.

- Led the club in organizing photography workshops and exhibitions.
- Coordinated with other student organizations for collaborative events.
- Organized a photography exhibition at the school, showcasing students' work and promoting the art of photography.

Paris, France
Sept 2018 – July 2019